THE ADJUNCTION-CONSTRUCTED PRODUCT ON HYPERCOHOMOLOGY IS THE CUP PRODUCT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement is a single sentence referring to the construction in the paragraph immediately before the lemma; that paragraph lies outside the counted span, and the construction is also written out inside the proof, so nothing load bearing is left outside.

Editorial note. The author's footnote isolating the sign hidden in the shift-tensor identification is part of the counted proof span and is reproduced as written.

Editorial note. The counted spans carry 1 navigation stub and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-cohomology/retained-preamble.tex lines 1-81), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-second-cup-equals-first is defined only in cohomology.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Adjunction between derived pullback and derived pushforward, and its counit. Derived Categories, the adjunction Lf^* and Rf_* with counit epsilon, used at proof lines 7222-7229. Supplies the description of the constructed product as the adjoint of a composite of counits.
- Shift-tensor identification for complexes of modules. More on Algebra, Item more-algebra-item-shift-tensor, invoked at proof line 7240. Supplies the identification A[-i] tensor A[-j] = A[-i-j] used to compare the two maps.
- Derived global sections as the derived pushforward to a point. Same chapter, the identification Rf_*(-) = RGamma(X, -) for f: X to a point, used at proof lines 7220-7221. Supplies the setting in which the cup product is defined.

Source reference. Original reference 0FNH: sources/stacks-chow/tags.txt, source lines 19567–19567. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *This construction gives the cup product.*

Proof. With $f : (X, \mathcal{O}_X) \to (pt, A)$ as above we have $Rf_*(-) = R\Gamma(X, -)$ and our map μ is adjoint to the map

$$Lf^*(Rf_*K \otimes^{\mathbf{L}}_A Rf_*M) = Lf^*Rf_*K \otimes^{\mathbf{L}}_{\mathcal{O}_X} Lf^*Rf_*M \xrightarrow{\epsilon_K \otimes \epsilon_M} K \otimes^{\mathbf{L}}_{\mathcal{O}_X} M$$

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where ϵ is the counit of the adjunction between Lf^* and Rf_*. If we think of ξ and η as maps $\xi : A[-i] \to R\Gamma(X, K)$ and $\eta : A[-j] \to R\Gamma(X, M)$, then the tensor $\xi \otimes \eta$ corresponds to the map[1]

$$A[-i-j] = A[-i] \otimes^{\mathbf{L}}_A A[-j] \xrightarrow{\xi \otimes \eta} R\Gamma(X, K) \otimes^{\mathbf{L}}_A R\Gamma(X, M)$$

By definition the cup product $\xi \cup \eta$ is the map $A[-i-j] \to R\Gamma(X, K \otimes^{\mathbf{L}}_{\mathcal{O}_X} M)$ which is adjoint to

$$(\epsilon_K \otimes \epsilon_M) \circ Lf^*(\xi \otimes \eta) = (\epsilon_K \circ Lf^*\xi) \otimes (\epsilon_M \circ Lf^*\eta)$$

However, it is easy to see that $\epsilon_K \circ Lf^*\xi = \tilde{\xi}$ and $\epsilon_M \circ Lf^*\eta = \tilde{\eta}$. We conclude that $\widetilde{\xi \cup \eta} = \tilde{\xi} \otimes \tilde{\eta}$ which means we have the desired agreement. □

[1]There is a sign hidden here, namely, the equality is defined by the composition

$$A[-i-j] \to (A \otimes^{\mathbf{L}}_A A)[-i-j] \to A[-i] \otimes^{\mathbf{L}}_A A[-j]$$

where in the second step we use the identification of More on Algebra, Item (0FNH) which uses a sign in principle. Except, in this case the sign is $+1$ by our convention and even if it wasn't $+1$ it wouldn't matter since we used the same sign in the identification $\mathcal{O}_X[-i-j] = \mathcal{O}_X[-i] \otimes^{\mathbf{L}}_{\mathcal{O}_X} \mathcal{O}_X[-j]$.